

ASSIGNMENT — 20 QUESTIONS

20
Total Marks

30 min
Time

MCQ
Type

2 — Closures & HOF
Day

Instructions

Har question ke liye sabse sahi option choose karo. Sab questions 1 mark ke hain.

Code-based questions mein mentally trace karo — console nahi dekho.

Answer key aakhiri page pe hai. 90%+ = Excellent. Below 70% = Topics dubara padhein.

Q1. [Closures] Closure kya hai?

- (A) Ek function jo apne arguments yaad rakhta hai
- (B) Ek function jo outer scope ke variables ko tab bhi access karta hai jab outer function return ho chuka ho
- (C) Ek immediately invoked function
- (D) Ek function jo doosre function ko return karta hai

Q2. [Closures] Ye code kya print karega? `function outer(){let x=1; return function inner(){console.log(x);}}
const fn=outer(); fn();`

- (A) undefined
- (B) ReferenceError
- (C) 1
- (D) null

Q3. [Closures] Closure mein 'private variable' pattern kyun kaam karta hai?

- (A) JavaScript mein private keyword hota hai
- (B) Closure outer scope ke variables ko encapsulate karta hai — bahar se accessible nahi
- (C) var se variables private ho jaate hain
- (D) IIFE se hi private variables bante hain

Q4. [Closure Loop] Ye buggy code kya print karega? `for(var i=0;i<3;i++){setTimeout(()=>console.log(i),0);}`

- (A) 0, 1, 2
- (B) 0, 0, 0
- (C) 3, 3, 3
- (D) undefined, undefined, undefined

Q5. [Closure Loop] Closure loop bug fix karne ke liye BEST approach kaunsi hai modern JS mein?

- (A) var ki jagah let use karo
- (B) IIFE use karo
- (C) bind() use karo
- (D) Array.from use karo

Q6. [map()] [1,2,3,4].map(x => x * 2) ka output?

- (A) [1,2,3,4]
- (B) [2,4,6,8]
- (C) [3,4,5,6]
- (D) 8

Q7. [filter()] [1,2,3,4,5].filter(x => x % 2 === 0) ka output?

- (A) [1,3,5]
- (B) [2,4]
- (C) [1,2,3,4,5]
- (D) false

Q8. [reduce()] [1,2,3,4].reduce((acc,n) => acc+n, 0) ka output?

- (A) 0
- (B) 10
- (C) 24
- (D) 1234

Q9. [map vs filter] map() aur filter() mein kya farq hai?

- (A) Koi farq nahi — dono same kaam karte hain
- (B) map() transform karta hai (same length), filter() select karta hai (smaller/equal length)
- (C) filter() transform karta hai, map() select karta hai
- (D) map() original array modify karta hai

Q10. [reduce()] reduce() mein initialValue miss karne se kya hoga agar array empty ho?

- (A) 0 return hoga
- (B) undefined return hoga
- (C) TypeError: Reduce of empty array with no initial value
- (D) null return hoga

Q11. [IIFE] IIFE ka full form kya hai?

- (A) Instantly Invoked Function Interface

- (B) Immediately Invoked Function Expression
- (C) Inner Invoked Function Expression
- (D) Immutable Invoked Function Instance

Q12. [IIFE] IIFE kyun use karte hain?

- (A) Performance improve karne ke liye
- (B) Global scope pollute na ho — private scope banana
- (C) Async code handle karne ke liye
- (D) Closures avoid karne ke liye

Q13. [Currying] `const add = a => b => a + b;` — `add(3)(4)` ka output?

- (A) 34
- (B) 7
- (C) undefined
- (D) NaN

Q14. [Currying] Currying ka main benefit kya hai?

- (A) Code faster run hota hai
- (B) Partial application — function ko pre-configure karke reuse kar sakte hain
- (C) Closures avoid ho jaate hain
- (D) Syntax short hoti hai

Q15. [Currying] `const withGST = rate => price => price+(price*rate/100);` — `withGST(18)(1000)` ka output?

- (A) 180
- (B) 1180
- (C) 18
- (D) 1000

Q16. [Memoization] Memoization kya hai?

- (A) Function ko faster likhna
- (B) Function ke results cache karna taaki same args pe dobara compute na ho
- (C) Function composition technique
- (D) Closures avoid karne ka pattern

Q17. [Memoization] Memoization ka cache kis concept pe based hai?

- (A) Prototype chain
- (B) Closure (private cache variable)
- (C) Event loop

(D) Promises

Q18. [Memoization] Memoization kab USE NAHI karna chahiye?

- (A) Jab function pure ho (same input = same output)
- (B) Jab function impure ho — jaise Date.now() ya random numbers
- (C) Fibonacci calculate karne ke liye
- (D) API response cache karne ke liye

Q19. [Composition] const pipe = (...fns) => x => fns.reduce((v,fn)=>fn(v),x); — ye kya karta hai?

- (A) Functions ko parallel mein run karta hai
- (B) Functions ko left-to-right chain mein apply karta hai
- (C) Functions ko right-to-left apply karta hai
- (D) Sab functions ek saath call karta hai

Q20. [HOF] Higher-Order Function kya hota hai?

- (A) Ek function jo sirf numbers pe kaam karta hai
- (B) Ek function jo doosre function ko argument mein leta hai ya return karta hai
- (C) Ek function jo globally available hota hai
- (D) Ek function jo sirf arrays pe kaam karta hai

Answer Key

Q1: (B)	Q2: (C)	Q3: (B)	Q4: (C)	Q5: (A)
Q6: (B)	Q7: (B)	Q8: (B)	Q9: (B)	Q10: (C)
Q11: (B)	Q12: (B)	Q13: (B)	Q14: (B)	Q15: (B)
Q16: (B)	Q17: (B)	Q18: (B)	Q19: (B)	Q20: (B)

Scoring Guide

20/20 (100%) — Outstanding! Day 3 (Prototypes & OOP) ke liye fully ready.

15-19 (75-95%) — Good! Weak topics (Closures / reduce) ek baar aur revise karo.

10-14 (50-70%) — Theory notes dobara padhein, exercises ek baar aur karo.

Below 10 (<50%) — Mentor se mil lo — Day 2 concepts Day 3 mein bhi aate hain.